

Brian Mugambi

Operational Data Analyst | Python, SQL, Revenue & Operations Analytics

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github.com/doinggbits/demurrage-revenue-engine

PROFESSIONAL SUMMARY

Computer Science graduate who builds data tools for operational problems: revenue leakage, billing disputes, data quality and risk scoring. Led the architecture of a tanker-waiting-fee (demurrage) billing platform for a Kenya Pipeline Company use case, and built analytics, dashboards and a document Q&A assistant in Python with automated tests. Open to Data Analyst and Systems Analyst roles (Nairobi, hybrid or remote).

SKILLS

Data & Analytics: Python, SQL (SQLite, PostgreSQL), Pandas, NumPy, data validation and quality checks, anomaly detection, KPI reporting, ROI and business-case analysis, Streamlit dashboards

Development: FastAPI, Java (Spring Boot), JavaScript, TypeScript, React, Next.js, REST APIs, MongoDB, pytest, LLM and RAG applications, AI-assisted development (Claude, GitHub Copilot)

Systems & Tools: Systems analysis, technical support, Linux system administration, network administration, Docker, Git/GitHub, Kafka, Redis

PROJECTS

KPC Revenue Assurance & Demurrage Platform | Lead Architect, team of 5 | Power Learn Project Capstone, 2026

- Led the architecture of a platform that turns gate and weighbridge records into verified tanker-waiting-fee invoices, replacing paper and spreadsheet billing. Built with FastAPI, Next.js and SQLite/PostgreSQL, run with Docker Compose.
- Built a 12-step rule-based charge calculator and a 6-rule compliance check that sends each charge to auto-approval, human review or rejection; every calculation carries a SHA-256 audit signature for auditors.
- Wrote SQL aggregations and an anomaly detector that flags depots running more than 30% above their turnaround baseline.
- On the project's demo dataset: 94.8% of charges captured and settlement in 4.2 days versus a 45 to 60 day manual cycle (illustrative data, not audited). Wrote the CFO/Board business case with ROI, payback and cautious-case scenarios.

ClimaCrop Intelligence (Kilimo-Smart) | Team member, team of 4 | Power Learn Project Capstone, 2026

- Contributed to a climate and credit analytics platform that turns 10 years of weather data (116 stations, 35 counties, 40 crops) into crop recommendations for farmers and one plain risk score for lenders.
- Delivered as two interfaces (Streamlit and Next.js) on one shared engine, released with 95 automated tests and a real data bug found and fixed before any farmer used it.

Ops Assistant: Document Q&A (RAG) Tool | Builder | Week 10 Project, 2026

- Built a Streamlit assistant that answers operators' questions from SOP PDFs, CSV incident logs and manuals, cites the file and page or row for every answer, and says "not found" instead of guessing.
- Runs offline or with an LLM API key stored in .env (no hardcoded secrets); 23 automated tests; reused for a second project by changing configuration only.

Beneficiary Identity-Resolution Dashboard | Builder | Hackathon Project

- Built a React/TypeScript and FastAPI dashboard that matches duplicate records using fuzzy matching (Jaro-Winkler, Levenshtein), scores anomalies with Isolation Forest, and logs every merge to an audit trail; added JWT login and a role-gated admin page.

Agency Banking Backend | Developer | Ongoing

- Working on a Java/Spring Boot microservices system (identity, agent, float, transaction and reporting services) with API gateway, Spring Cloud Config, Kafka, Redis and PostgreSQL, deployed with Docker Compose; building a reporting service with a Kafka-fed audit trail.

EXPERIENCE

Information Technology Assistant | Kenya National Bureau of Statistics | May 2023 to August 2023

EDUCATION

Bachelor of Computer Science, The Co-operative University of Kenya | September 2020 to December 2025

Power Learn Project (PLP) | Capstone and hackathon programme, 2026